

SwiftChain Logistics Inc.

Supply Chain Delivery Intelligence System — Project Brief

Issued: April 20, 2026 | Priority: HIGH | Deadline: 1 Week

1. Company Overview

SwiftChain Logistics Inc. is a mid-sized third-party logistics (3PL) provider operating across the United States. The company manages end-to-end shipment fulfilment for B2B, B2C, and Government clients through a network of **10 warehouses, 200 routes, and 5 carrier partners**. Despite strong order volume, on-time delivery performance has declined to **48.9 %** — well below the industry benchmark of 85 %. Leadership has commissioned this analytics project to identify root causes and recommend corrective actions.

2. Business Problem

The Operations Director needs data-driven answers to:

- Why are deliveries getting delayed and which factors contribute most?
- Which routes consistently underperform on delivery timelines?
- Which warehouses are responsible for the highest delay rates?
- What is the financial impact of delays on shipping cost and order value?
- Which carrier has the worst on-time delivery performance?
- How do weather and traffic conditions affect delay rates?
- Where should we focus improvement efforts for maximum impact?

3. Dataset Overview — swiftchain_logistics_450k.csv

Company: SwiftChain Logistics Inc.

Total Records: 450,800 rows (including ~800 injected duplicates)

Date Range: January 2021 – December 2024

Columns: 28

File: swiftchain_logistics_450k.csv

Column	Type	Description
Shipment_ID	Text	Unique shipment identifier
Order_Date	Date	Date order was placed
Promised_Delivery_Date	Date	Expected delivery date
Actual_Delivery_Date	Date	Actual delivery date (3% NULL — in transit)
Delay_Days	Integer	Actual minus Promised (negative = early)
Delay_Status	Text	Early / On Time / Slightly Delayed / Delayed / Severely Delayed
On_Time_Flag	Integer	1 = On Time or Early, 0 = Delayed

Origin_City / State	Text	Shipment origin location
Origin_Region	Text	East / West / South / Central
Destination_City/State	Text	Delivery destination
Destination_Region	Text	East / West / South / Central
Route_ID	Text	200 unique routes (20 flagged as bad routes)
Distance_Miles	Integer	Route distance in miles
Warehouse_ID	Text	10 warehouses (WH-001 to WH-010)
Warehouse_City/State	Text	Warehouse location
Carrier	Text	FedEx / UPS / DHL / USPS / OnTrac
Ship_Mode	Text	Air / Express / Ground / Standard
Product_Category	Text	8 categories (Electronics, Furniture, etc.)
Product_Weight_kg	Decimal	Shipment weight in kilograms
Order_Value	Decimal	Dollar value of the order
Shipping_Cost	Decimal	Actual shipping cost charged
Customer_Segment	Text	B2B / B2C / Government
Customer_ID	Text	8,000 unique customers
Weather_Condition	Text	Clear / Rain / Snow / Storm / Fog (5% NULL)
Traffic_Condition	Text	Low / Medium / High / Severe (4% NULL)

4. Real-World Data Issues (you must handle these)

Issue	Details	Action Required
NULL values	13,697 rows missing Actual_Delivery_Date 22,707 NULL Weather 18,220 NULL Traffic	Impute or exclude depending on analysis
Duplicate rows	~800 exact duplicate shipment records injected	Identify and remove
Outlier delays	~1,300 rows with 30-45 day delays (strikes/customs)	Flag and document
Negative delay days	Some shipments arrived early (delay = -1 to -3)	Keep — valid business data
Mixed delay categories	5 delay statuses — some overlap in reporting	Define clear thresholds

5. KPIs to Build

#	KPI Name	Formula	Page
1	On-Time Delivery Rate	On Time Orders / Total Orders x 100	1
2	Average Delay Days	AVG(Delay_Days) WHERE Delay_Days > 0	1
3	Total Shipments	COUNT(Shipment_ID)	1
4	Total Shipping Cost	SUM(Shipping_Cost)	1
5	Delay Cost Impact	Delayed Orders x AVG(Shipping_Cost)	1
6	Avg Order Value	SUM(Order_Value) / COUNT(Shipment_ID)	1
7	Route Efficiency Score	On Time % per Route	2
8	Carrier On-Time Rate	On Time % per Carrier	2
9	Ship Mode Performance	On Time % per Ship Mode	2
10	Distance vs Delay	Correlation: Distance Miles vs Delay Days	2
11	Warehouse On-Time Rate	On Time % per Warehouse	3
12	Weather Delay Impact	AVG Delay Days by Weather Condition	3
13	Traffic Delay Impact	AVG Delay Days by Traffic Condition	3
14	Monthly Delay Trend	Delay Rate % by Month	3
15	Seasonal Delay Pattern	On-Time Rate by Quarter	3

6. Dashboard Blueprint (3 Pages)

Page 1 — Executive Overview

- KPI Cards: Total Shipments, On-Time Rate, Avg Delay Days, Total Cost
- Monthly Delay Trend — Line Chart
- On-Time Rate by Region — Bar Chart
- Delay Rate by Carrier — Bar Chart

- Shipments by Customer Segment — Donut Chart
- Year and Region slicers

Page 2 — Route and Carrier Performance

- Top 10 Worst Routes by Delay Rate — Bar Chart
- Carrier Comparison — Clustered Bar (On-Time % vs Avg Delay)
- Ship Mode Performance — Bar Chart
- Distance vs Delay Days — Scatter Plot
- Route Performance Table — sortable
- Carrier and Ship Mode slicers

Page 3 — Warehouse and Root Cause Analysis

- Warehouse Performance Ranking — Bar Chart
- Weather Condition Impact — Bar Chart
- Traffic Condition Impact — Bar Chart
- Delay Heatmap: Month vs Region — Matrix Visual
- Product Category Delay Analysis — Bar Chart
- Warehouse and Year slicers

7. Deliverables and Timeline

Day	Task	Deliverable	Status
1	Setup + Excel Exploration	Data Dictionary + EDA Notes	Pending
2	Python EDA + Cleaning	cleaned_swiftchain.csv + .ipynb	Pending
3	Python Advanced Analysis	KPI outputs + charts saved	Pending
4	SQL Normalization + Queries	15+ SQL KPI queries .sql file	Pending
5	Power BI Data Model	Star schema + relationships	Pending
6	Power BI Dashboard Build	3-page dashboard .pbix	Pending
7	GitHub + README	Public repo with all files	Pending

8. Manager Expectations

- **Tell a story with data** — every chart must answer a specific business question.
 - **Every insight needs a recommendation** — don't just show the problem, suggest the fix.
 - **Cross-validate all numbers** — Python totals must match SQL totals must match Power BI totals.
 - **Clean, commented code** — SQL scripts and Python notebooks must be readable by anyone.
 - **Professional GitHub** — README must explain the project clearly with screenshots.
 - **Quantify the impact** — always express findings in dollar or percentage terms.
 - **Root cause first** — do not stop at 'DHL is slow'; explain WHY DHL is slow.
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9. Hidden Insights in the Data (find these!)

The following patterns are embedded in the dataset. Your analysis should surface all of them:

#	Finding	Expected Impact
1	WH-002, WH-005, WH-008 responsible for 60%+ of severe delays	Warehouse process overhaul
2	DHL on-time rate ~28% below FedEx	Carrier contract review
3	Q4 (Nov-Dec) delay rate 60% higher than Q1-Q3	Seasonal capacity planning
4	Storm weather adds avg 5 extra delay days	Weather-based routing
5	20 bad routes account for majority of delays	Route optimisation
6	Electronics + Pharma have highest delay rates	Priority handling SLA
7	Ground shipping 3x more delayed than Air	Mode shift recommendation
8	~800 duplicate records inflating KPIs	Data quality remediation
